When LLMs Hallucinate: Examining the Effects of Erroneous Feedback in Math Tutoring Systems

Transparent Changes to the Preregistration

The preregistration can be found at
https://osf.io/5yk7s/overview?view_only=34bcbb666baa44cd816bad48e382f5fa

1. Description of change: Instead of excluding participants that took less than 20 minutes, we exclude cases where participants completed less than 4 steps in the acquisition phase
    a. Rationale: Our analysis is based on engagement per feedback instance (i.e., the data is in long format with one row per question) and we believe that this approach more accurately excludes participants based on their actual engagement with the experiment. This approach further enables us to exclude participants who skipped the experiment and consequently did not take part in the intervention.
    b. Effect of change on study results: None expected.
2. Description of change: Missing data is handled with Full Information Maximum Likelihood (FIML; Enders, 2001) instead of Multiple imputation
    a. Rationale: Multiple imputation is not always necessary for non-hierarchical data but introduces problems with moderated regession models (Graham, 2009; Lüdtke et al., 2007), which can easier be handled with FIML.
    b. Effect of change on study results: None expected.